

Preliminary Progress Review (PPR) Report

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Project Title: “SafeLAN”: Adaptive Secure File System for LAN environment

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1. Introduction

1.1 Purpose

The purpose of this Preliminary Progress Review (PPR) document is to formally report and communicate the current status of the SafeLAN project at this stage of its development. This document provides a structured overview of the research background, methodology, implementation progress and planned activities, including ongoing tasks during the initial phases of the project. It ensures that the project remains aligned with the approved research objectives. It also serves to demonstrate alignment with the approved research plan.

The intended audience for this PPR includes the project supervisors and the relevant assessment panel responsible for assessing project progress. This document is designed to support informed feedback and guidance ensuring that the research project continues to progress in a controlled, feasible, and academically rigorous manner.

1.2 Definitions, Acronyms, and Abbreviations

LAN	Local Area Network
2FA	Two-factor Authentication
API	Application Programming Interface
MediaPipe	A framework for building multimodal machine learning pipelines
OTP	One Time Password
OpenCV	Open-Source Computer Vision Library
SVM	Support Vector Machine
UI	User Interface
PoA	Proof of Authority
MITM	Man-in-the-Middle Attack
SMEs	Small-to-Medium Enterprises

1.3 Overview

This Preliminary Progress Review (PPR) document is structured to present the current progress and planned direction of the SafeLAN research project in a clear and logical manner. Following this introductory section, the report includes a review of relevant literature, the formulation of the research problem and the definition of the research objectives.

The report then describes the proposed research methodology and identifies the sources for test data and analysis. This is followed by a discussion of the anticipated benefits of the research, the defined scope and expected research outcomes and any identified research constraints. Finally, PPR outlines the project plan and schedule to illustrate the planned progression of the research activities.

2. Literature review

Despite extensive research in the domain of secure file systems, a significant gap remains in the development of offline, LAN-specific security solutions that operate independently of internet connectivity and do not rely solely on default operating system-level permissions. Most existing secure file systems primarily rely on static authentication mechanisms such as usernames, passwords and operating system-level access permissions, which provide only point-of-entry verification. These mechanisms are unable to verify whether the individual entering valid credentials is the legitimate account owner in the LAN system. Also they are unable to assess the context of access, such as the trustworthiness of the device, location or time of access, particularly in LAN environments where centralized identity services and real-time external verification are unavailable. Prior research has consistently demonstrated that static authentication methods are ineffective in detecting insider threats, credential misuse, unauthorized user behaviour and post-login attacks such as session hijacking and physical takeover scenarios [1] [2]. Since authorized users typically operate within permitted privilege boundaries, traditional access control models fail to identify malicious actions occurring after successful authentication. These limitations are especially critical in LAN-based systems that operate without internet connectivity, where security enforcement is limited to local mechanisms [3].

Researchers have proposed continuous authentication systems using machine learning-based behavioural anomaly detection to address these shortcomings. These systems analyze user behaviour through mouse movement patterns, system interaction characteristics and biometric signals like face recognition, to verify identity throughout an active session. Studies report improved detection accuracy for post-login threats when behavioural biometrics and anomaly detection techniques are employed. However, the majority of these solutions are designed for mobile, cloud or internet-connected environments, leaving offline LAN-based continuous authentication insufficiently explored.

In parallel, advancements in computer vision technologies, including OpenCV and MediaPipe have enabled gesture-based interaction and authentication using standard webcams. Although gesture recognition has been applied in human-computer interaction, its usage as a secure authentication mechanism to view files in offline desktop environments remains limited. Existing literature shows that two-factor authentication (2FA) is commonly implemented using mechanisms such as one-time passwords (OTPs), mobile authenticator applications and biometric methods to improve security in file viewing of secure systems [4]. However, most existing 2FA solutions rely on continuous internet connectivity, third-party services or additional hardware. Also, there is limited research focusing on gesture-based authentication approaches within offline or desktop application environments to limit file viewing feature. Furthermore, research on audit

logging increasingly recommends blockchain-based approaches to ensure immutability and accountability [3]. However, most existing solutions depend on cloud-based or public blockchain infrastructures, which are unsuitable for restricted LAN environments due to connectivity and privacy constraints.

Overall, the literature reveals a clear lack of integrated, offline, LAN-focused security systems that combine adaptive authentication, continuous identity verification, controlled file access and immutable audit logging within a single framework. This research addresses this gap by proposing SafeLAN, a secure LAN-based file system that unifies behavioural biometrics-based adaptive authentication along with a trust-score, continuous authentication, gesture-controlled file viewing and blockchain-based auditing without reliance on internet connectivity.

2.1 Identification and significance of the problem

Despite extensive research in secure file systems, existing solutions remain inadequate for offline Local Area Network (LAN) environments that operate without internet connectivity and centralized cloud-based security services. The following key problem areas highlight the limitations of traditional LAN-based file systems and justify the need for the proposed SafeLAN framework.

1. Limitations of credential-based authentication

Traditional LAN file systems primarily rely on static authentication mechanisms such as usernames, passwords, and operating system-level permissions. These mechanisms verify only the correctness of the entered credentials, not the legitimacy of the individual typing them. As a result, any user possessing valid credentials can successfully authenticate, regardless of whether the access is authorized or malicious. This creates a critical vulnerability in environments where credentials may be shared, stolen, or misused by insiders.

2. Inability to detect insider threats and credential misuse

Insider threats represent a significant security challenge in LAN environments, as malicious actions are often performed using valid credentials and within assigned privilege boundaries. Traditional access control systems are not designed to distinguish between legitimate users and insiders misusing authorized access. Consequently, unauthorized data access, data exfiltration and policy violations can occur without triggering security alerts, particularly in offline LAN environments.

3. Absence of context-aware trust evaluation

Conventional LAN file systems do not consider contextual factors such as the device being used, the location of access or the time of access when granting permissions. In offline environments where real-time external verification is unavailable, access decisions are made without assessing the trustworthiness of the access context. This lack of trust evaluation increases exposure to risks

such as compromised devices, unauthorized system usage outside permitted hours, and unauthorized physical access.

4. Lack of continuous authentication after login

In Local Area Network (LAN) environments, security mechanisms primarily rely on one-time authentication at login, assuming that the authenticated user remains legitimate throughout the session. There are no mechanisms to continuously verify user identity during active sessions. This assumption creates a critical vulnerability, enabling post-login attacks such as session hijacking, man-in-the-middle (MITM) and physical hijacking of unattended systems. These attacks allow adversaries to exploit trusted sessions without detection, where attackers exploit an already authenticated session without needing to re-enter credentials.

5. Inadequate control over secure file viewing

Existing file systems provide limited control over how sensitive files are accessed and viewed after authentication. Users can often copy, modify or misuse files without additional verification or restrictions. There is only a limited exploration of hand gesture-based authentication as a biometric factor in two-factor authentication systems in LAN-based file systems when viewing files. Unlike conventional biometric methods such as fingerprint or facial recognition, this approach proposes using dynamic hand gesture-based digit representation for OTP input. The significance of this problem lies in investigating a novel, touchless biometric modality that does not require specialized hardware and can potentially improve both security and usability in LAN-based desktop applications.

6. Vulnerability of centralized audit logging mechanisms

Traditional audit logs in LAN-based systems are typically stored in centralized locations and can be altered or deleted by privileged users particularly by administrative users or attackers. This lack of immutability reduces accountability and complicates forensic analysis following security incidents. The problem is that the lack of secure, tamper-proof and role-aware audit mechanisms for LAN-based file access systems. The significance of this problem lies in the absence of lightweight blockchain solutions that support dual visibility, where administrators can access complete audit trails while standard users are restricted to viewing only their own access records.

The combined impact of these limitations highlights the absence of a comprehensive, offline, LAN-based security framework that integrates adaptive authentication, continuous user verification, controlled file access and tamper-proof auditing. As organizations such as small-to-medium enterprises, educational institutions, healthcare facilities and government offices increasingly rely on LAN-based systems with restricted internet access, the need for a unified and robust security solution becomes critical. Addressing these challenges is essential to improving data confidentiality, integrity, accountability and trust within LAN environments.

3. Research Question

This research aims to investigate how an integrated, offline, LAN-based security framework can enhance authentication, access control, and accountability beyond traditional static security mechanisms. To achieve this objective, the study is guided by the following primary research question along with other sub questions that will be addressed in the main project.

The primary research question is;

How can an integrated, offline security framework combining adaptive authentication, continuous behavioural verification, controlled file access, and tamper-proof audit logging enhance protection against insider threats and unauthorized access in LAN-based file systems?

1. Adaptive authentication and trust scoring

- How can behavioural biometrics, such as keystroke dynamics and a trust score computed with contextual login attributes (device identity, access location and time of access) be integrated for adaptive authentication in LAN-based file systems?
- Can a trust score-based adaptive authentication mechanism provide stronger resistance against insider threats and credential misuse compared to traditional static username and password authentication methods?

2. Continuous authentication and post-login threat detection

- How can continuous, behaviour-driven authentication mechanisms validate user legitimacy beyond initial login in offline LAN environments?
- To what extent can machine learning-based behavioural anomaly detection and face recognition techniques identify and mitigate post-login threats such as session hijacking, physical workstation takeover, and man-in-the-middle (MITM) attacks while maintaining acceptable usability and low false-positive rates?

3. Secure file viewing through gesture-based authentication

- How can a hand gesture-based biometric approach be integrated with one-time password (OTP) verification to develop a secure and usable two-factor authentication system for file viewing option in desktop applications, particularly in offline or limited-connectivity environments

4. Blockchain-based audit logging in LAN environments

- How can a dual-visibility permissioned blockchain architecture using a Proof of Authority (PoA) consensus mechanism be designed and integrated into a LAN-based file access system to ensure secure, transparent, and role-based audit logging?
- Can blockchain-based audit logs improve accountability, traceability, and forensic reliability compared to traditional centralized logging mechanisms in offline environments?

4. Research Objectives

The primary objective of the SafeLAN project is to design, develop, and evaluate a secure, offline, LAN-based file system that mitigates insider threats, credential misuse, and post-login attacks by integrating adaptive authentication, continuous verification, controlled file access, and tamper-proof auditing without reliance on internet connectivity.

To achieve this overarching goal, the specific research objectives are as follows:

- To develop a behavioural biometrics-based (keystroke) adaptive authentication mechanism along with a trust score, based on contextual data of the user for Local Area Network file system without internet dependency.
- To identify whether the user who is typing the credentials is the actual user.
- To bridge the security gap in LAN file system authentication modules by introducing a real-time trust score-based, biometric authentication layer to reduce the attack surface associated with relying solely on static passwords.
- To design, implement, and evaluate a machine learning-based continuous authentication system for Local Area Network (LAN) environments to detect and mitigate post-login attacks such as session hijacking, man-in-the-middle (MITM), and physical hijacking using behavioral anomaly detection and face recognition.
- To design, implement, and evaluate a machine learning-based continuous authentication system for Local Area Network (LAN) environments that uses behavioral anomaly detection and face recognition to identify and mitigate post-login attacks, including session hijacking, man-in-the-middle (MITM), and physical hijacking, while maintaining high detection accuracy and low false positives.
- Design and develop a prototype two-factor authentication system for desktop applications that integrates hand gesture-based biometric input with one-time password (OTP) verification.
- Investigate the feasibility and effectiveness of using computer vision techniques, specifically MediaPipe and OpenCV, for real-time recognition of hand gesture-based digit input in an offline authentication environment.
- Design permissioned blockchain architecture suitable for LAN environments using a Proof of Authority (PoA) consensus mechanism.
- Implement a dual-visibility blockchain model that enforces role-based access to audit data for administrators and standard users.
- Integrate the blockchain system with the SafeLAN file access control framework to record file access, modification, and sharing events as immutable transactions.
- Develop a blockchain backend service capable of block creation, validation, and integrity verification.
- Design and implement a graphical block explorer interface to visualize blockchain records and support audit verification.

5. Research methodology

This research adopts a design-based experimental methodology to develop and evaluate SafeLAN, a comprehensive secure desktop application for Local Area Network (LAN) environments that integrates multiple advanced security mechanisms. The system combines behavioural biometrics including keystroke dynamics and facial features with machine learning based anomaly detection to assess user legitimacy and identify deviations from normal behavior. In addition, hand gesture-based One-Time Password (OTP) authentication provides a privacy preserving, offline alternative to typed codes and facial biometrics, reducing risks associated with keylogging, spoofing, and deepfake attacks. SafeLAN also implements a permissioned blockchain framework to record all file access events, including encryption, decryption, viewing, and permission changes, ensuring immutable, tamper-resistant audit logging.

A Proof of Authority (PoA) consensus mechanism and role-based access control enable efficient block validation and dual visibility for standard users and administrators. The system is tested in a controlled LAN environment under simulated attack scenarios, such as session hijacking, man-in-the-middle attacks, and physical credential compromise, and is evaluated using metrics including detection accuracy, false positive rate, authentication success, system response time, and blockchain transaction integrity. This integrated methodology validates the feasibility, effectiveness, and usability of combining keystroke biometrics, gesture-based authentication, behavioral anomaly detection, and blockchain audit logging in a unified, user-friendly, offline LAN-based file access control system.

6. Sources for test data and analysis

Data Collection Procedures:

Test data will be collected within a controlled Local Area Network (LAN) environment using volunteer participants interacting with the SafeLAN system. Behavioural metrics including keystroke dynamics, mouse movements, system interaction logs, and hand gesture inputs for OTP authentication will be recorded during routine file operations as well as during simulated attack scenarios such as session hijacking, physical hijacking, and man-in-the-middle (MITM) attacks. File access activities including creation, modification, viewing, and permission changes will also be logged. Hand gesture data will be captured using a pre-trained MediaPipe Hand Landmarker model to enable offline gesture recognition under varying lighting conditions and camera qualities. All data will be anonymized and labeled to protect privacy and ensure compliance with ethical standards.

Data Analysis Methods:

The collected data will be analyzed using a combination of descriptive and quantitative methods to evaluate the performance, security, and usability of SafeLAN as an integrated system. Behavioral anomaly detection models will be assessed using metrics such as accuracy, precision, recall, and false positive rate. Hand gesture authentication will be evaluated for recognition accuracy, authentication success rate, and response time. Blockchain-based audit logs will be analyzed for integrity, completeness, role-based visibility enforcement, and tamper resistance, with transaction validation performed through PoA authority nodes and verified via the block explorer GUI. Together, these analyses provide a comprehensive assessment of the system's effectiveness in securing LAN-based file access while maintaining usability and operational efficiency.

7. Anticipated benefits

The proposed research is expected to significantly enhance the security of LAN-based file access systems by introducing an adaptive, behaviour-aware authentication framework that operates independently of internet connectivity. By leveraging behavioural biometrics, specifically keystroke dynamics, the system enhances its ability to accurately identify whether the individual entering credentials is the legitimate user. This approach reduces reliance on static password-based authentication solely, which is highly vulnerable to credential compromise, replay attacks, and insider misuse. The integration of a trust score mechanism, computed using behavioural and contextual user data, strengthens access control decisions by providing an additional security layer beyond traditional authentication methods.

By introducing continuous and behaviour-aware authentication mechanisms beyond the traditional one-time login, the system reduces the risk of unauthorized access caused by session hijacking and physical or logical compromise. Real-time behavioural monitoring supported by machine learning techniques enables the early detection of anomalous activities, thereby strengthening intrusion prevention while minimizing false positives and preserving normal user workflows.

In addition, the integration of a permissioned blockchain-based audit logging mechanism ensures tamper-resistant, transparent and traceable records of file access activities within the LAN. The use of a lightweight consensus mechanism such as Proof of Authority enables efficient and reliable audit logging suitable for private network environments.

Overall, the proposed solution provides a cost-effective, privacy-preserving, and user-friendly security framework for LAN-based file systems, offering both practical benefits to organizations and valuable contributions to research in adaptive authentication and secure offline systems.

8. Scope and specified deliverables / expected research outcome

The scope of this research encompasses the development of SafeLAN, a comprehensive desktop application designed to provide secure file access and management within Local Area Network (LAN) environments. SafeLAN integrates multiple security mechanisms to enhance authentication, authorization, and auditability without relying on internet connectivity. The system incorporates behavioral biometrics, such as keystroke dynamics, to verify user identity and compute a trust score based on contextual and behavioral data, reducing reliance on static passwords and mitigating risks associated with session hijacking, man-in-the-middle attacks, and physical credential compromise. It also supports machine learning-based anomaly detection to identify unusual or potentially malicious behavior, improving detection accuracy while minimizing false positives.

To strengthen authentication further, SafeLAN introduces hand gesture-based One-Time Password (OTP) input as a privacy-preserving alternative to typed codes or facial biometrics, ensuring secure access even under conditions vulnerable to spoofing or keylogging attacks. All file operations, including creation, modification, viewing, and permission changes, are logged through a permissioned blockchain that provides immutable, tamper-resistant audit trails, enforced via a Proof of Authority consensus mechanism with role-based visibility for administrators and standard users. The expected deliverables include a fully integrated SafeLAN prototype demonstrating adaptive trust-based authentication, gesture-based OTP verification, anomaly detection, secure file management and blockchain-enabled audit logging, along with an evaluation report analyzing system performance, security effectiveness, and usability.

The anticipated research outcome is a proof-of-concept application that validates the feasibility, reliability, and practical applicability of combining behavioral biometrics, machine learning, privacy-preserving authentication, and blockchain auditability to create a secure, user-friendly, and scalable LAN-based file access system.

9. Research constraints

The study is conducted within a controlled Local Area Network (LAN) environment, which limits evaluation to offline and small-scale network settings and does not account for system performance or security behaviour in large-scale network infrastructures. The research is further constrained by limited dataset availability, hardware and resource constraints, and user privacy considerations, all of which influence the scope of data collection, system training and experimental evaluation.

Additionally, the system need to be designed, implemented and tested without internet connectivity, thereby excluding the use of cloud-based services, public blockchain platforms and online code-generation or validation tools. Consequently, the blockchain component operates as a permissioned and private distributed ledger within the LAN, with the primary focus placed on validating functional correctness, audit immutability and access traceability rather than achieving large-scale decentralization or global consensus.

10. Project plan or schedule

The SafeLAN project follows a phased approach including literature review, system design, data collection, model development, system integration, testing and documentation. Each phase is scheduled in accordance with the academic timeline to ensure feasibility and timely completion. The project plan emphasizes modularity, offline operation and progressive validation to support seamless integration of all components into a unified system.

1. Project initiation and literature review

This phase focuses on establishing the theoretical and technical foundation of the project.

- Conduct an in-depth literature review on:
 - LAN-based secure file systems
 - Comparison of static and adaptive authentication mechanisms
 - Behavioural biometrics and keystroke dynamics
 - Continuous authentication and post-login threat detection
 - Gesture-based authentication
 - Blockchain-based audit logging in restricted environments
- Identify research gaps, system requirements, and constraints specific to offline LAN environments
- Finalized functional and non-functional requirements, research objectives, research questions, and overall SafeLAN architecture

The above is already completed.

2. Integrated security component development

This phase focuses on developing and validating the proposed security mechanisms as independent yet interoperable modules, without initial dependency on the base SafeLAN application.

2.1. Adaptive Authentication and Trust Score Mechanism

- Collect behavioural biometric data using keystroke dynamics (dwell time and flight time)
- Capture contextual attributes such as device identity, LAN IP address, and time of access
- Train user-specific One-Class SVM models to establish normal behavioural boundaries
- Design and implement a trust score mechanism combining behavioural and contextual indicators
- Evaluate resistance to credential misuse and insider threats during authentication

2.2 Continuous Authentication and Post-Login Threat Detection

- Implement continuous behavioural monitoring during active sessions
- Integrate face recognition for identity verification using local camera input
- Apply machine learning-based anomaly detection to identify session hijacking, MITM attempts, and physical takeover scenarios
- Optimize detection thresholds to maintain usability and low false-positive rates

2.3 Secure File Viewing Using Gesture-Based Authentication

- Develop gesture recognition models using computer vision techniques
- Integrate gesture-based verification as a second authentication factor for file access
- Enforce file viewing restrictions within an isolated desktop environment
- Validate usability and effectiveness in offline conditions

2.4 Blockchain-Based Audit Logging

- Design a permissioned blockchain using a Proof of Authority (PoA) consensus model
- Define role-based visibility and access control for audit records
- Log authentication events, trust scores, file access activities, and security alerts
- Validate immutability, transparency, and performance within a LAN environment

3. Base SafeLAN application development and integration

This phase integrates all developed security components into a functional LAN-based file system.

- Develop the core SafeLAN desktop application
- Establish LAN-based communication without internet dependency
- Implement features like secure file storage, retrieval, session management and access workflows
- Integrate adaptive authentication, continuous monitoring, gesture-based file access, and blockchain auditing into the base system
- Conduct end-to-end testing in a controlled LAN environment
- Perform performance, security, and usability evaluations

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